

1.HTML Structure

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. It defines the content and layout of a webpage by organizing elements such as headings, paragraphs, images, links, tables, and forms.

HTML is interpreted by web browsers, which render the webpage for users.

Basic HTML Document Structure

Every HTML document follows a standard structure.

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>My First Web Page</title>

  </head>

  <body>

    <h1>Welcome to HTML</h1>

    <p>This is my first webpage.</p>

  </body>

</html>
```

2.<!DOCTYPE html>

The DOCTYPE (Document Type Declaration) is a declaration placed at the very beginning of an HTML document. It tells the web browser which version of HTML the document uses so that the browser can render the webpage correctly.

For HTML5, the declaration is: <!DOCTYPE html>

Although it is written like an HTML tag, `<!DOCTYPE html>` is not an HTML element. It is simply an instruction to the browser.

3.<head> Element

The `<head>` element is a container that holds metadata (information about the webpage). The content inside the `<head>` is not displayed on the webpage. Instead, it provides information that helps browsers, search engines, and other web services understand and process the document.

The `<head>` element is placed between the `<html>` and `<body>` elements.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <!-- Metadata goes here -->
```

```
</head>
```

4.</body> Element

The `<body>` element is the section of an HTML document that contains all the visible content displayed in a web browser. Everything that users see and interact with on a webpage such as headings, paragraphs, images, links, tables, forms, videos, and buttons is placed inside the `<body>` element.

The `<body>` element is located after the `<head>` element and is a required part of every HTML document.

5.Meta Tags

Meta tags are HTML elements that provide metadata (information about a webpage). They are placed inside the `<head>` section of an HTML document and are not displayed on the webpage. Instead, they provide information to web browsers, search engines, and other web services about the document.

Meta tags are commonly used to specify the character encoding, viewport settings, page description, author information, and search engine keywords.

Examples:

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<meta name="description" content="An introductory HTML tutorial.">
```

```
<meta name="author" content="Ups Shah">
```

```
<meta name="keywords" content="HTML, CSS, JavaScript">
```

6.Semantic HTML

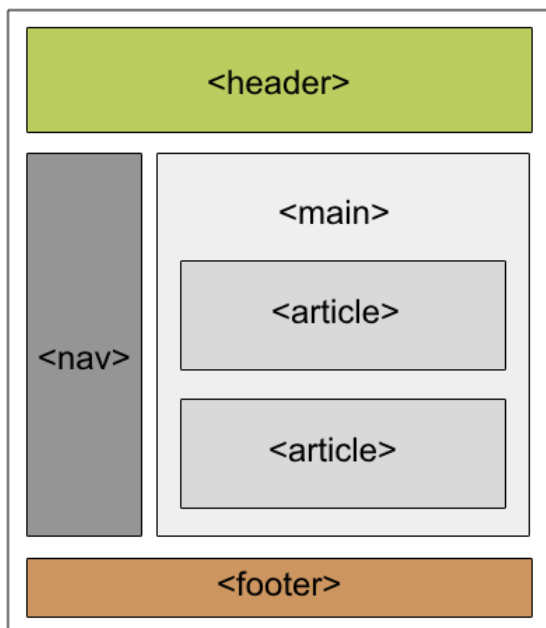
Semantic HTML refers to the use of HTML elements that clearly describe the meaning and purpose of the content they contain. Instead of using generic elements such as `<div>` or `<span>` for everything, semantic elements indicate the role of different sections of a webpage.

Semantic HTML makes web pages more readable, accessible, and search engine friendly.

Common Semantic HTML Elements are :

Element	Purpose
<code><header></code>	Represents the introductory section or header of a webpage or section.
<code><nav></code>	Contains navigation links.
<code><main></code>	Represents the primary content of the webpage.
<code><section></code>	Groups related content into logical sections.
<code><article></code>	Represents self-contained content, such as a blog post or news article.

<code><aside></code>	Contains supplementary or sidebar content related to the main content.
<code><footer></code>	Represents the footer of a webpage or section.
<code><figure></code>	Groups media such as images, diagrams, or illustrations.
<code><figcaption></code>	Provides a caption for the content inside a <code><figure></code> .
<code><address></code>	Contains contact information for the author or organization.



7. Headings

HTML headings are used to define the titles and subheadings of a webpage. They help organize content into a logical structure, making it easier for users to read and for search engines and assistive technologies to understand the page hierarchy.

HTML provides six levels of headings, ranging from `<h1>` to `<h6>`.

- `<h1>` is the highest and most important heading.
- `<h6>` is the lowest and least important heading.

8.Paragraph

A paragraph in HTML is used to display a block of text. It is created using the <p> (paragraph) element, which automatically adds spacing before and after the text to improve readability.Paragraphs are commonly used to display articles, descriptions, explanations, and other textual content on a webpage. Syntax:

```
<p>This is a paragraph.</p>
```

9.Lists

HTML lists are used to organize and display related items in a structured format. Lists improve the readability of web pages by presenting information as a sequence of items rather than as plain text.

HTML provides three types of lists:

1. **Unordered List ()** – Displays items with bullets.
2. **Ordered List ()** – Displays items with numbers or letters.
3. **Description List (<dl>)** – Displays terms and their descriptions.

1. Unordered List ()

An unordered list displays items using bullet points. It is used when the order of items is not important.

Syntax

```
<ul>
```

```
<li>Item 1</li>
```

```
<li>Item 2</li>
```

```
<li>Item 3</li>
```

```
</ul>
```

Example

```
<h2>Programming Languages</h2>
```


HTML

CSS

JavaScript

React

Output

- HTML
- CSS
- JavaScript
- React

2. Ordered List ()

An ordered list displays items in a specific order using numbers by default.

Syntax

First Item

Second Item

Third Item

Example

<h2>Steps to Create a Webpage</h2>

Create an HTML file.

Add the HTML structure.

`<li>Save the file.</li>`

`<li>Open it in a browser.</li>`

`</ol>`

Output

1. Create an HTML file.
2. Add the HTML structure.
3. Save the file.
4. Open it in a browser.

Ordered List Types

The type attribute changes the numbering style.

Type	Example
1	1, 2, 3 (Default)
A	A, B, C
a	a, b, c
I	I, II, III
i	i, ii, iii

Example

`<ol type="A">`

`<li>HTML</li>`

```
<li>CSS</li>
```

```
<li>JavaScript</li>
```

```
</ol>
```

Output

A. HTML

B. CSS

C. JavaScript

3. Description List (<dl>)

A description list is used to display terms and their corresponding descriptions.

It consists of:

- <dl> = Description List
- <dt> = Description Term
- <dd> = Description Details

Syntax

```
<dl>
```

```
<dt>Term</dt>
```

```
<dd>Description</dd>
```

```
</dl>
```

Example

```
<dl>
```

```
<dt>HTML</dt>
```

```
<dd>The standard markup language for creating web pages.</dd>
```

```
<dt>CSS</dt>
```

```
<dd>Used to style HTML documents.</dd>
```


`<dt>JavaScript</dt>`

`<dd>Used to add interactivity to web pages.</dd>`

`</dl>`

Output

HTML

The standard markup language for creating web pages.

CSS

Used to style HTML documents.

JavaScript

Used to add interactivity to web pages.

Nested Lists

A list can contain another list inside one of its items.

Example

`<ul>`

`<li>Frontend`

`<ul>`

`<li>HTML</li>`

`<li>CSS</li>`

`<li>JavaScript</li>`

`</ul>`

`</li>`

`<li>Backend`

`<ul>`

```
<li>Node.js</li>

<li>Express.js</li>

</ul>

</li>

</ul>
```

10.Links

HTML links, also known as hyperlinks, are used to connect one webpage to another, navigate to different sections of the same page, download files, or open email addresses. Links allow users to navigate between web pages by clicking on text, images, or other HTML elements. HTML links are created using the `<a>` (anchor) element.

Syntax

```
<a href="URL">Link Text</a>
```

Syntax Explanation:

Attribute/Element	Description
<code><a></code>	Defines a hyperlink (anchor element).
<code>href</code>	Specifies the destination URL of the link.
Link Text	The clickable text displayed to the user.

11.Images

Images are used to display visual content such as photographs, illustrations, icons, and logos on a webpage. They make web pages more attractive, informative, and engaging. In HTML, images are inserted using the `<img>` element. Unlike most HTML elements, the `<img>` element is a self-closing (void) element, meaning it does not require a closing tag.

Syntax

```

```

Syntax Explanation:

Attribute	Description
src	Specifies the path or URL of the image.
alt	Provides alternative text if the image cannot be displayed. It also improves accessibility for screen readers.

12. Table

HTML tables are used to organize and display data in rows and columns. They are commonly used to present structured information such as schedules, student records, product lists, and reports. A table consists of rows (<tr>), columns (created using <th> or <td>), and an optional caption.

Basic Table Structure and the main elements used to create a table are:

Element	Purpose
<table>	Defines the table.
<tr>	Defines a table row.
<th>	Defines a table header cell.
<td>	Defines a table data cell.
<caption>	Provides a title for the table.

Syntax :

```
<table>
  <tr>
    <th>Header 1</th>
    <th>Header 2</th>
  </tr>

  <tr>
    <td>Data 1</td>
    <td>Data 2</td>
  </tr>
</table>
```

13. HTML Forms

HTML forms are used to collect input from users and send that information to a server for processing. Forms are commonly used for login pages, registration forms, contact forms, search boxes, surveys, and feedback forms.

The <form> element acts as a container for various input controls, such as text fields, radio buttons, checkboxes, drop-down lists, and submit buttons.

Basic Form Structure

```
<!DOCTYPE html>

<html lang="en">

<head>

  <title>HTML Form</title>

</head>

<body>

  <h2>Student Registration Form</h2>

  <form action="/register" method="post">

    <label for="name">Name:</label><br>

    <input type="text" id="name" name="name"><br><br>

    <label for="email">Email:</label><br>

    <input type="email" id="email" name="email"><br><br>

    <input type="submit" value="Register">

  </form>

</body>

</html>
```

14.Accessibility

Accessibility (often abbreviated as a11y) is the practice of designing and developing websites so that everyone, including people with disabilities, can access and use them effectively.

Accessibility ensures that users with visual, hearing, motor, or cognitive impairments can navigate, understand, and interact with web content using assistive technologies such as screen readers, keyboards, and voice recognition software.

Why is Accessibility Important?

Accessibility provides several benefits:

- Makes websites usable by everyone.
- Improves the user experience.
- Supports users with disabilities.
- Enhances Search Engine Optimization (SEO).
- Helps websites comply with accessibility standards and legal requirements.

Common Accessibility Practices:

1. Use Semantic HTML

Semantic elements help browsers and screen readers understand the structure of a webpage.

```
<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
</nav>

<main>
  <section>
    <h2>Latest News</h2>
  </section>
</main>
<footer>
  Copyright 2026
```

</footer>

2. Add Alternative Text to Images

Always provide an alt attribute for images.

```

```

If the image cannot be displayed, the browser or a screen reader uses the alternative text.

3. Associate Labels with Form Controls

Use the <label> element with the for attribute.

```
<label for="email">Email</label>
```

```
<input  
  type="email"  
  id="email"  
  name="email">
```

This allows screen readers to correctly identify form fields.

4. Use Meaningful Link Text

Avoid generic link text like "Click Here".

```
<a href="notes.pdf">Click Here</a>
```

```
<a href="notes.pdf">  
  Download HTML Notes  
</a>
```

Descriptive links help users understand the destination before clicking.

5. Maintain Proper Heading Order

Use headings in a logical hierarchy.

```
<h1>HTML Tutorial</h1>
```

```
<h2>Introduction</h2>
```

```
<h3>History of HTML</h3>
```

Do not skip heading levels unnecessarily.

15. Git Command

What is Git?

Git is a distributed version control system (VCS) used to track changes in source code during software development. It allows developers to manage different versions of a project, collaborate with others, and restore previous versions if needed.

Git records the history of changes made to files and enables multiple developers to work on the same project simultaneously.

Benefits:

- Tracks changes made to files.
- Maintains a history of project versions.
- Enables collaboration among multiple developers.
- Allows reverting to previous versions if necessary.
- Integrates with platforms such as GitHub, GitLab, and Bitbucket.

git init

The `git init` command initializes a new Git repository in the current project directory. It creates a hidden `.git` folder that contains all repository metadata, configuration files, and version history. After initialization, Git begins tracking changes made within the project.

git clone

The `git clone` command creates a local copy of an existing remote repository, such as one hosted on GitHub, GitLab, or Bitbucket. In addition to downloading the project files, it also retrieves the complete commit history, branches, and repository configuration, enabling developers to work on the project locally.

git add

The `git add` command stages new or modified files before they are committed. Staging allows developers to select specific changes that should be included in the next commit, providing greater control over version history.

git commit

The `git commit` command records the staged changes as a permanent snapshot in the repository's history. Each commit represents a specific version of the project and is accompanied by a descriptive message that summarizes the changes made.

git status

The git status command displays the current state of the repository. It identifies modified files, staged files, untracked files, and the current branch, allowing developers to review the repository before performing additional Git operations.

git log

The git log command displays the commit history of the repository. It provides information such as commit identifiers, authors, commit dates, and commit messages, allowing developers to review the evolution of the project over time.

git push

The git push command uploads local commits to a remote repository. It synchronizes the local repository with the remote version, making recent changes available to other collaborators and ensuring the remote repository remains up to date.

git pull

The git pull command retrieves the latest changes from a remote repository and merges them into the local repository. It is commonly used before starting new work to ensure the local project is synchronized with the most recent updates made by other contributors.